

38 In the α -particle scattering experiment, a beam of α -particles is aimed at a thin gold foil.

Most of the α -particles go straight through or are deflected by a small angle. A very small proportion are deflected by more than 90° , effectively rebounding towards the source of the α -particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

- A** Most of the atom is empty space.
- B** Most of the mass of an atom is concentrated in the nucleus.
- C** The nucleus contains both protons and neutrons.
- D** The nucleus is charged.